Smart City Environmental Monitoring and Sustainability Analytics System



Meet the Team



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Background

Smart cities in India are unable to monitor their environment and sustainability as a whole, leading to several environmental issues and low quality of life.

Traffic departments record vehicle flow, pollution control boards record air quality, meteorological agencies record weather, and utilities record energy use. Because each dataset has its own format, sampling rate and owner, planners rarely see how these factors interact. For example, a PM2.5 spike may be blamed on traffic when low wind speed or high energy demand is also at play.

Without an integrated view, cities cannot easily:

- identify which factors drive pollution at different times of day and seasons
- measure sustainability with a single comparable indicator
- forecast pollution early enough to issue warnings or adjust traffic and energy policy

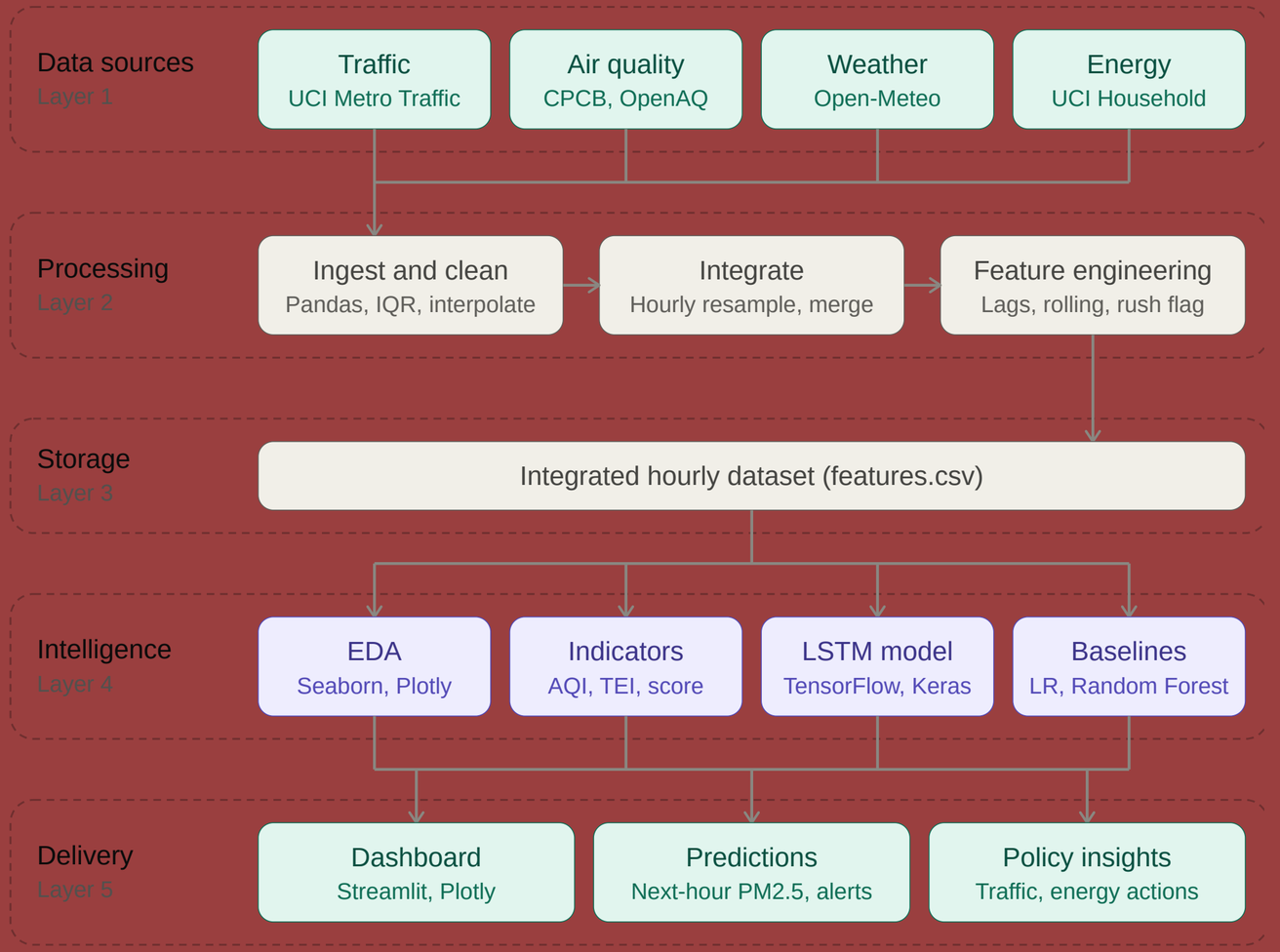
Research Objectives

- To analyze relationships among traffic, weather and pollutants, and to compute sustainability indicators (AQI category distribution, Traffic-Emission Index and a 0-100 Sustainability Score).
- To develop an LSTM-based TensorFlow model that predicts next-hour PM2.5 and compare it with persistence, Linear Regression and Random Forest baselines.





System Architecture



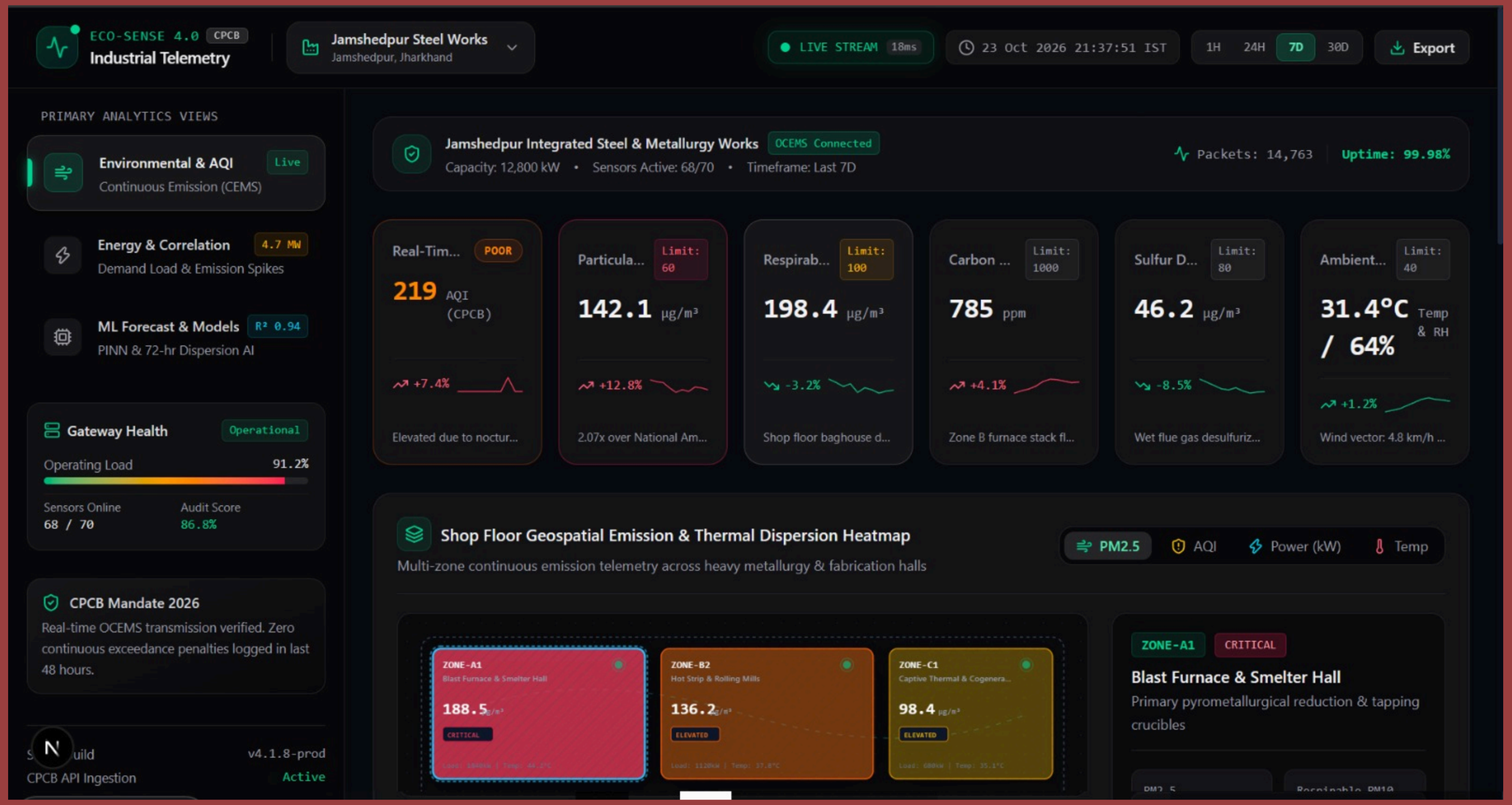
Our Unique Features

One pipeline that fuses Delhi’s air quality, weather and grid electricity demand into a single daily sustainability view, and tests how much each data source improves PM2.5 forecasts.

- **It connects three domains that are usually treated separately.** Real grid demand and a 0-100 sustainability score sit alongside the pollution forecast, so one dashboard shows air quality, energy load and weather stress together.
- **Provides planning insights.** Our application not only predicts and displays the results. but also provides live feedback for smart city planning.

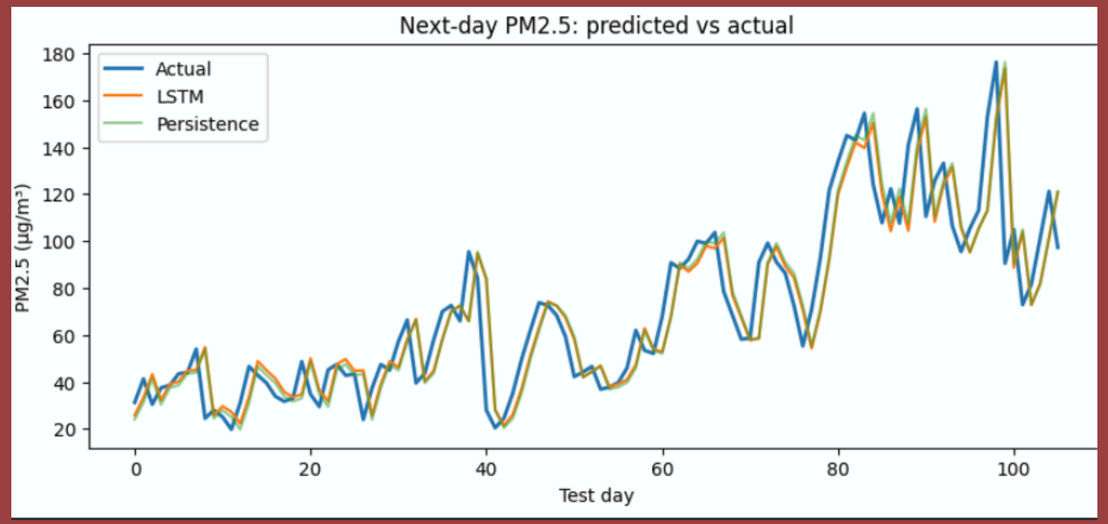


Our Unique Features



Our Dashboard with Representative Data

Conclusion



	RMSE	MAE	R2
All features	17.659	12.749	0.768
Without weather	17.634	12.743	0.768
Without energy	17.549	12.679	0.771
Without other pollutants	17.388	12.755	0.775
PM2.5 history only	16.873	12.422	0.788

On the test period (16 August to 30 November 2024), all models performed within about 4% of each other. Persistence reached an RMSE of 17.94 µg/m³, Ridge regression 17.43, Random Forest 17.20 and the LSTM 17.66 (R^2 of 0.76 to 0.78). The LSTM beat persistence but not the Random Forest, which is expected given only about 480 training days. The feature ablation showed that PM2.5 history carries most of the predictive information, and that weather and energy inputs add little to a next-day forecast on this much data. Their main value lies in the sustainability indicators and the seasonal insight that Delhi's electricity demand peaks in summer while its pollution peaks in winter.

The main limitations are the small sample, a single location, air quality data that is probably modelled, and the exclusion of traffic.

These insights will prove helpful in city planning and similar models can be built and extended for other smart cities.

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Complete Code

https://github.com/lord-of-the-strings/Smart-City-Monitoring-System

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